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On the Connectivity of Large-Scale Hybrid Wireless Networks

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***Abstract*—Many real systems are hybrid networks which include infrastructure nodes in multi-hop wireless networks, such as sinks in sensor networks and mesh routers in mesh networks. However, we have very little understanding of network connectivity in such networks. Therefore, in this paper, we consider hybrid networks denoted by $H(\alpha,\beta)$ with ad hoc nodes and base stations and prove how base stations can improve the connectivity of ad hoc nodes in *subcritical phase*, that is, the ad hoc node density, λ_α is lower than the critical density λ_α^c. We first study the impact of density of base stations, λ_β on the connectivity, and find that with the existence of a positive density of base stations which have the same transmission range as ad hoc nodes, i.e., $\lambda_\beta > 0$, the number of connected ad hoc nodes is $\Theta(n)$ with probability nearly 1, where n is the number of ad hoc nodes. However, the size of connected ad hoc component scales linearly with λ_β with probability nearly 1 when λ_β is lower than $c_1(\lambda_\alpha)$. We then study the impact of transmission range of base stations, r_β on the connectivity, and find the additional benefit of enlarging r_β to enhance the connectivity of ad hoc nodes.**

I. Introduction

Connectivity is a basic feature in multihop wireless networks in which a wireless node must establish a communication path to other nodes in order to successfully transmit or receive information. To this end, there have been extensive studies on the connectivity of wireless ad hoc networks and sensor networks, in which a large number of wireless nodes, either mobile or static, communicate with each other in a multi-hop fashion [1]–[3].

However, large-scale pure ad hoc networks rarely exist in real systems, because the per-node throughtput can diminish to zero [4], [5]. The most widely used systems are in fact hybrid networks which include infrastructure nodes in multi-hop wireless networks, such as sinks in sensor networks and mesh routers in mesh networks. In recent years, there have been considerable studies of such networks [6], [7], which have mainly focused on the *capacity* of hybrid networks. For example, Liu *et al*. in [6] investigate the capacity of hybrid networks, in which n ad hoc nodes are uniformly and independently located in a unit disk, but m base stations are placed on a regular grid within the area of the network. Their results show that the benefit of adding base stations on capacity is insignificant if m grows asymptotically slower than $\sqrt{n}$. But if m grows asymptotically faster than $\sqrt{n}$, the capacity increases linearly with the number of base stations, which provides an effective improvement over pure ad hoc networks.

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One interesting problem, yet remains unsolved, is to what extent the base stations improve the network connectivity, even though it appears to be intuitive that base stations can surely improve network connectivity. As an example, Fig. 1 shows that two connected components of ad hoc nodes, Subnet-A_1 and Subnet-A_2 with size 6 and 3 respectively, are separate before two base stations, B_1 and B_2, are added in the network. Nevertheless, after B_1 and B_2 are added, Subnet-A_1 and Subnet-A_2 are connected together and thus form a larger connected component.

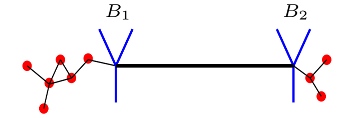

Fig. 1. A hybrid network with two base stations and ad hoc nodes.

Therefore, in this paper, our objective is to study the connectivity in hybrid networks, and we focus specifically on the case that ad hoc node density, λ_α is lower than the critical density λ_α^c, under which the connectivity in pure ad hoc network is "bad". To study this problem, we first consider a pure ad hoc network which is in subcritical phase, and utilize numerical simulations to examine the upper bound of *origin stretch* $\mathcal{OS}(\lambda_\alpha)$, which is defined as the largest Euclidean distance between the origin and the node in the connected component containing the origin. Here, the upper bound of $\mathcal{OS}(\lambda_\alpha)$, denoted by $c_0(\lambda_\alpha)$, is defined as the value by which $\mathcal{OS}(\lambda_\alpha)$ is bounded with probability nearly 1.

Then we proceed to study a hybrid network $H(\alpha,\beta)$ in which the transmission ranges of ad hoc nodes (α-nodes) r_α is normalized to unit 1 and base stations (β-nodes) are multiples of this unit, denoted by r_β. First, we consider unit transmission range of β-nodes, i.e., $r_\beta = 1$, and study the impact of density of base stations, λ_β on the connectivity. Through the analysis, we find out that with the existence of a positive density of base stations $\lambda_\beta > 0$, the number of connected ad hoc nodes is $\Theta(n)$ with probability nearly 1, where n is the number of ad hoc nodes. However, the size of connected ad hoc component scales linearly with λ_β that is lower than $c_1(\lambda_\alpha)$ with probability nearly 1, which demonstrates a tremendous

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benefit of using base stations to enhance the connectivity of ad hoc nodes.

Further, we analyze the impact of transmission range of base stations, r_β on the connectivity. Our results indicate that by increasing r_β from unit to $k_0 c_2(\lambda_\alpha)+1$ ($k_0 \in \mathbb{N}$), with a smaller density of β-nodes, β-giant component can reach a larger size, which demonstrates an additional benefit of enlarging r_β to improve the connectivity of α-nodes.

The rest of the paper is organized as follows. In Section II, we introduce the network model and provide upper bound of origin stretch, $c_0(\lambda_\alpha)$ for different density of ad hoc nodes. In Section III, we present the impact of density and transmission range of base stations, followed by the conclusions in Section IV.

II. Upper Bound of Origin Stretch $\mathcal{OS}(\lambda_\alpha)$

A. Hybrid Network Model

There are two types of nodes in the hybrid network $H(\alpha,\beta)$. The first set consists of ad hoc nodes which are called α-nodes. A set of n α-nodes $\mathcal{A}=\{\alpha_1,\alpha_2,\ldots,\alpha_n\}$ are randomly deployed at locations $A=\{a_1,a_2,\ldots,a_n\}$ in a square region D with side length d, by use of Poisson distribution with density λ_α. This square region D is also called a continuum box [8]. We assume that every α-node has the same transmission power and signal receiving capability, and thus has the same transmission range denoted as r_α. That is, for any two α-nodes $\alpha_i,\alpha_j \in \mathcal{A}$, there is an undirectional link between them iff $\|a_i-a_j\| \le r_\alpha$, where $\|\cdot\|$ denotes the Euclidean distance. Without loss of generality, we normalize r_α to unit 1 throughout the paper. Then we denote *random geometric graph* (RGG) $G(\mathcal{H}_{\lambda_\alpha},1)$ as the ensemble of all α-nodes and the links between them. It is well known from continuum percolation theory [8], [9], that there exists a critical density $0<\lambda_\alpha^c<\infty$ in $G(\mathcal{H}_{\lambda_\alpha},1)$ such that: 1) for $\lambda_\alpha>\lambda_\alpha^c$, there exists a unique connected component containing $\Theta(n)$ nodes a.a.s. (supercritical phase), and 2) for $\lambda_\alpha<\lambda_\alpha^c$, the connected component contains a finite number of nodes a.s. (subcritical phase). Until recently, the exact value of λ_α^c is still unknown although some bounds have been given in the literatures. The analytical abound $0.696<\lambda_\alpha^c<3.372$ is given in [9], while the simulation bound $1.434<\lambda_\alpha^c<1.438$ is given in [10]. In this paper, we assume the density of α-nodes, λ_α is lower than λ_α^c.

The second set consists of base stations which are called β-nodes. A set of m β-nodes $\mathcal{B}=\{\beta_1,\beta_2,\ldots,\beta_m\}$ (with density $\lambda_\beta=\frac{m}{d^2}$) are placed regularly at the locations $B=\{b_1,b_2,\ldots,b_m\}$ $([-\frac{d}{2}+\frac{d}{2\sqrt{m}}+\frac{d}{\sqrt{m}}i,-\frac{d}{2}+\frac{d}{2\sqrt{m}}+\frac{d}{\sqrt{m}}j],0\le i,j\le\sqrt{m}-1)$ in the region D as depicted in Fig. 2. The β-nodes are connected by high bandwidth wirelines and act only as relay nodes for routing and forwarding the packets from and to the α-nodes. We assume that β-nodes are much more powerful than α-nodes that are power-constrained [4], and therefore have a transmission range of $r_\beta \ge r_\alpha = 1$. That is, for any pair of α-node $\alpha_i\in\mathcal{A}$ and β-node $\beta_j\in\mathcal{B}$, there is an undirectional link between them iff $\|a_i-b_j\|\le r_\beta$.

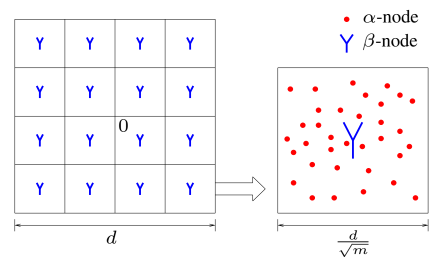
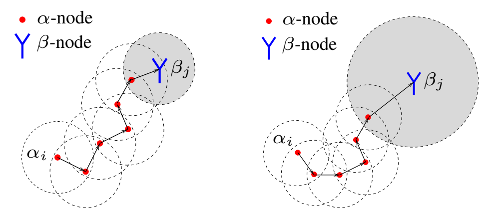


Fig. 2. The layout of a hybrid network.



(a) $r_\beta=1$ (b) $r_\beta=2$

Fig. 3. Data flow from an α-node to a β-node.

Figs. 3(a) and 3(b) demonstrate an example of data flow from an α-node α_i to a β-node β_j in the case $r_\beta=1$ and $r_\beta=2$ respectively. In both cases, there exists one α-node, which is in the transmission range of β_j, and can communicate with α_i in a multi-hop fashion. Therefore, the data sent from α_i can reach β_j through the relay of α-nodes. As a result, the hybrid network $H(\alpha,\beta)$, is defined as the ensemble of all α-nodes, β-nodes, and links (wireline and wireless) between them.

B. Definitions

Definition 1 (β-Component): In hybrid network $H(\alpha,\beta)$, the *β-component* denoted by $\mathcal{C}_{(\alpha,\beta)}$ is defined as the connected component containing all β-nodes, and a subset of α-nodes that are connected by the β-nodes. Note that, we assume all β-nodes are connected via broadband wireline, that is, if a connected component contains one β-node, it will contain all β-nodes, of which the number are constant. Thus, we define *β-component size* $|\mathcal{C}_{(\alpha,\beta)}|$ as the number of α-nodes in $\mathcal{C}_{(\alpha,\beta)}$. For example, as depicted in Fig. 1, there are two β-nodes in hybrid network, connecting two separate subsets together to form a larger connected component. The connected component including B_1, B_2, Subset-A_1 and Subset-A_2 is the β-component.

Definition 2 (β-Giant Component): In hybrid network $H(\alpha,\beta)$, the *β-giant component*, is the β-component with size $|\mathcal{C}_{(\alpha,\beta)}|=\Theta(n)$.

Definition 3 (Origin Stretch): Let $G(\mathcal{H}_{\lambda_\alpha,0},1)=G(\mathcal{H}_{\lambda_\alpha},1)\cup\{0\}$, and $\mathcal{C}_0$ be the connected component in $G(\mathcal{H}_{\lambda_\alpha,0},1)$ containing the origin 0. The *origin stretch* of $G(\mathcal{H}_{\lambda_\alpha,0},1)$ is defined as $\mathcal{OS}(\lambda_\alpha)=\max_{\alpha_i\in\mathcal{C}_0}\|\alpha_i-0\|$, where $\|\alpha_i-0\|$ is the Euclidean distance between α_i and the origin 0. For example, as depicted in Fig. 4, origin stretch $\mathcal{OS}(\lambda_\alpha)$ is the Euclidean distance between α_i and the origin 0.

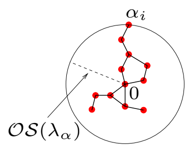


Fig. 4. Origin stretch in pure ad hoc network.

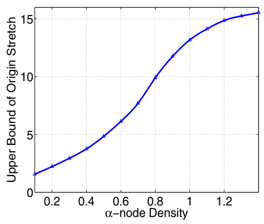


Fig. 5. The upper bound of origin stretch in pure ad hoc networks.

C. Upper Bound of Origin Stretch $\mathcal{OS}(\lambda_\alpha)$

Our object in this paper, is to study how β-nodes can improve the connectivity of α-nodes, and investigate what are steps. First, we start from a special case, that is, α-nodes only and find the upper bound of origin stretch $\mathcal{OS}(\lambda_\alpha)$ for different given α-node density, to understand the coverage of α-nodes. Here, the upper bound of origin stretch $\mathcal{OS}(\lambda_\alpha)$, denoted by $c_0(\lambda_\alpha)$, is defined as the value by which the origin stretch is bounded with probability nearly 1.

We now use numerical simulations to compute $c_0(\lambda_\alpha)$. In the simulation, 10000 α-nodes are distributed independently and randomly with a uniform distribution to approximate a Poisson point process. The α-node density λ_α is set between 0.1 and 1.4 to guarantee that the pure ad hoc network is in subcritical phase. Specifically, the upper bound of origin stretch $\mathcal{OS}(\lambda_\alpha)$, $c_0(\lambda_\alpha)$, is set as the value by which the origin stretch is bounded with probability over 99%.

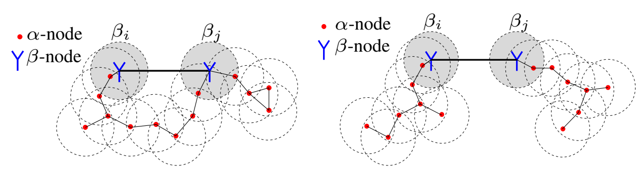
Fig. 5 illustrates the value of $c_0(\lambda_\alpha)$ for different α-node density. We can see from the figure as α-node density increases from 0.1 to 1.4, $c_0(\lambda_\alpha)$ increases accordingly. This result can be explained in the following way: as α-node density λ_α increases, the average number of α-nodes in the transmission range (mean degree) of any give α-node increases, and so for any given α-node in the network, it can be connected with other α-nodes with a higher probability. Therefore, $\mathcal{C}_0$ becomes larger and stretch farther away with a higher probability. In addition, we notice that $c_0(\lambda_\alpha)$ does not scale linearly with α-node density λ_α. When λ_α is small, $c_0(\lambda_\alpha)$ increases fast with the addition of λ_α; however, when λ_α approaches the critical density λ_α^c, the increasing rate of $c_0(\lambda_\alpha)$ decreases significantly.

III. How do β-Nodes improve the connectivity of α-nodes?

From previous discussions, we have observed that β-nodes can improve network connectivity. An interesting question is that how many β-nodes need to be added to a network. Therefore, we first evaluate *β-nodes* utilization in network connectivity and introduce a new concept called "*efficient use of β-nodes*".

A. Efficient Use of β-nodes in Hybrid Network $H(\alpha,\beta)$

As depicted in Fig. 6(a), two β-nodes, β_i and β_j, are connected through the relay of a number of α-nodes. Let I be the number of α-nodes connected "directly" by β_i. Here, we say α-nodes are connected "directly" by a β-node, iff these α-nodes can be connected by the β-node even if we remove all other β-nodes. Similarly, let J be the number of α-nodes connected "directly" by β_j. For the particular case shown in the figure, we can easily find that $I = 9$ and $J = 13$. However, the total number of α-nodes, which are connected "directly" by β_i and β_j, is $13 < I + J$. In Fig. 6(b), β_i and β_j cannot be connected through the relay of α-nodes. Then the total number of α-nodes, which are connected "directly" by β_i and β_j, is $14 = I + J$. Apparently, in the later case, the two β-nodes are more efficiently used. Therefore, we define that all β-nodes in hybrid network $H(\alpha,\beta)$ are efficiently used, iff any two β-nodes cannot be connected through the relay of α-nodes.



(a) Unefficient use of β-nodes (b) Efficient use of β-nodes

Fig. 6. Unefficient use vs. efficient use of β-nodes ($r_\beta = 1$).

B. Impact of β-node density, λ_β

The impact of β-nodes on the connectivity of a hybrid network is two-fold: the effect of node density λ_β and the effect of transmission range r_β on network connectivity. In this section, we consider unit β-nodes transmission range, i.e., $r_\beta = 1$, and study the effect of β-node density λ_β on network connectivity.

Theorem 1: For unit transmission range of β-nodes, if the β-node density satisfies $\lambda_\beta < c_1(\lambda_\alpha)$, all β-nodes in hybrid network $H(\alpha,\beta)$ are efficiently used with probability nearly 1, where $c_1(\lambda_\alpha) = \frac{1}{c_2^2(\lambda_\alpha)}$ and $c_2(\lambda_\alpha) = c_0(\lambda_\alpha) + 1$.

Proof: As we define before, $c_0(\lambda_\alpha)$ is the the value by which origin stretch $\mathcal{OS}(\lambda_\alpha)$ is bounded with probability nearly 1. Then, given that event E_{bound} that α-nodes connected by a β-node β_i "directly" is bounded within the disk centered at β_i with radius $c_0(\lambda_\alpha)$ happens, if the Euclidean distance between the adjacent β-nodes is larger than $c_0(\lambda_\alpha) + 1$, all β-nodes in hybrid network $H(\alpha,\beta)$ are efficiently used. Otherwise, if two adjacent β-nodes β_i and β_j can be connected through the relay of α-nodes, there must exists one α-node α_s that is within the unit disk centered at β_j and can be connected by β_i through the relay of α-nodes. This is equivalent to say, α_s is not within the disk centered at β_i with radius $c_0(\lambda_\alpha)$, and thus contradicts with the condition that event E_{bound} happens. Then from $\frac{d}{\sqrt{m}} > c_0(\lambda_\alpha) + 1$, we know $\lambda_\beta = \frac{m}{d^2} < \frac{1}{(c_0(\lambda_\alpha)+1)^2}$. $\square$

Consider the case that all β-nodes in hybrid network $H(\alpha,\beta)$ are efficiently used. If we ignore the wireline connection between the β-nodes, each connected component containing a β-node is an independent connected component, which can be viewed as $\mathcal{C}_0$ in $G(\mathcal{H}_{\lambda_\alpha,0},1)$. Then in hybrid network $H(\alpha,\beta)$, β-component size is the summation of the number of α-nodes connected by each β-node "directly". In other word, let k_i ($i \in \{1,2,\cdots,m\}$) be the number of α-nodes that are connected by β_i "directly". Then in hybrid network $H(\alpha,\beta)$, β-component size is equal to $|\mathcal{C}_{(\alpha,\beta)}| = \Sigma_{i=1}^m k_i$. To compute the value of $\Sigma_{i=1}^m k_i$, we give the following lemma.

Lemma 1: Let $|\mathcal{C}_0|$ be the number of α-nodes in $\mathcal{C}_0$ of $G(\mathcal{H}_{\lambda_\alpha,0},1)$. When $G(\mathcal{H}_{\lambda_\alpha,0},1)$ is in subcritical phase, the mean of $|\mathcal{C}_0|$: $E(|\mathcal{C}_0|)$, and the variance of $|\mathcal{C}_0|$: $\mathrm{Var}(|\mathcal{C}_0|)$, are both finite.

Proof: From Theorem 9.7 in [8], when $G(\mathcal{H}_{\lambda_\alpha,0},1)$ is in subcritical phase, there exist constants $\mu > 0$, $g_0 > 0$ such that $\mathrm{Prob}(|\mathcal{C}_0| \geq g) \leq e^{-\mu g}$ when $g \geq g_0$. Hence, we have $E(|\mathcal{C}_0|) = \Sigma_{g=1}^{\infty} g \cdot \mathrm{Prob}(|\mathcal{C}_0| = g) = \Sigma_{g=1}^{g_0} g \cdot \mathrm{Prob}(|\mathcal{C}_0| = g) + \Sigma_{g=g_0}^{\infty} g \cdot \mathrm{Prob}(|\mathcal{C}_0| = g) \leq \Sigma_{g=1}^{g_0} g \cdot \mathrm{Prob}(|\mathcal{C}_0| = g) + \Sigma_{g=g_0}^{\infty} g \cdot e^{-\mu g}$, and $E(|\mathcal{C}_0|^2) = \Sigma_{g=1}^{\infty} g^2 \cdot \mathrm{Prob}(|\mathcal{C}_0| = g) \leq \Sigma_{g=1}^{g_0} g^2 \cdot \mathrm{Prob}(|\mathcal{C}_0| = g) + \Sigma_{g=g_0}^{\infty} g^2 \cdot e^{-\mu g}$. By using the ratio test criteria, we know both $\Sigma_{g=g_0}^{\infty} g \cdot e^{-\mu g}$ and $\Sigma_{g=g_0}^{\infty} g^2 \cdot e^{-\mu g}$ converge. Thus, both $E(|\mathcal{C}_0|)$ and $E(|\mathcal{C}_0|^2)$ are finite, and further $\mathrm{Var}(|\mathcal{C}_0|) = E(|\mathcal{C}_0|^2) - E^2(|\mathcal{C}_0|)$ is finite. □

Based on the results in Lemma 1, we will investigate the variance of the independent and identically distributed (i.i.d.) random variable k_i. In fact, k_i is 1 less than $|\mathcal{C}_0|$, since $|\mathcal{C}_0|$ accounts the origin 0 as an α-node. Thus, it is easy to know both the mean of k_i and the variance of k_i are also finite, with $E(k_i) = E(|\mathcal{C}_0|) - 1$ and $\mathrm{Var}(k_i) = \mathrm{Var}(|\mathcal{C}_0|)$. Then, let $\eta(\lambda_\alpha) = E(k_i)$ be the mean of k_i for a given λ_α, i.e., $\eta(\lambda_\alpha) = \mathrm{E}(|\mathcal{C}_0|) - 1$. Then, we will have the following theorem.

Theorem 2: With a density of $0 < \lambda_\beta < c_1(\lambda_\alpha)$ β-nodes in hybrid network $H(\alpha,\beta)$, β-giant component will emerge with probability nearly 1. Further, β-giant component size scales linearly with λ_β as $|\mathcal{C}_{(\alpha,\beta)}| = \frac{\eta(\lambda_\alpha)n}{\lambda_\alpha}\lambda_\beta + o(n)$ with probability nearly 1.

Proof: From Theorem 1, when $\lambda_\beta < c_1(\lambda_\alpha)$, all β-nodes in the hybrid network are efficiently used with probability nearly 1. Then, as we discuss before, if all β-nodes in the hybrid network are efficiently used, β-component size can be computed by $|\mathcal{C}_{(\alpha,\beta)}| = \Sigma_{i=1}^m k_i$, in which k_i ($i \in \{1,2,\cdots,m\}$) are i.i.d. random variables with finite mean $\eta(\lambda_\alpha)$ and finite variance. When $\lambda_\beta > 0$, $m \to \infty$, using the *Weak Law of Large Numbers* [11], we know $|\mathcal{C}_{(\alpha,\beta)}|$ converges to $m\eta(\lambda_\alpha) \pm o(m)$ almost surely. Since $m = \lambda_\beta d^2 = \frac{\lambda_\beta n}{\lambda_\alpha}$, we have $|\mathcal{C}_{(\alpha,\beta)}| = \frac{\eta(\lambda_\alpha)n}{\lambda_\alpha}\lambda_\beta + o(n) = \Theta(n)$, which indicates β-giant component emerges. □

Theorem 2 indicates that β-component size scales linearly with λ_β with probability nearly 1, when $0 < \lambda_\beta < c_1(\lambda_\alpha)$. Further, the result in Theorem 2 also shows that with a density of λ_β ($0 < \lambda_\beta < \lambda_\alpha^c - \lambda_\alpha$) β-nodes in hybrid network, β-giant component, which contains $\Theta(n)$ α-nodes, will emerge with probability nearly 1. Thus, adding base stations in the pure ad hoc network will bring tremendous benefit on the connectivity of α-nodes.

C. Impact of β-nodes transmission range, r_β

As the transmission range of β-nodes, r_β increases from unit, more α-nodes will be inside the disk centered at a single β-node with radius r_β, that is, a single β-node can connect together more connected component of α-nodes, which are separate without the help of β-nodes. In this section, we will study the impact of r_β on the connectivity of α-nodes. First, we give a lemma as follows.

Lemma 2: In hybrid network $H(\alpha,\beta)$ for $r_\beta > 1$, if the β-node density satisfies $\lambda_\beta < c_3(\lambda_\alpha)$, all β-nodes in the network are efficiently used with probability nearly 1, where $c_3(\lambda_\alpha) = \frac{1}{(r_\beta + c_2(\lambda_\alpha))^2}$.

Proof: Similar to the proof in Theorem 1, when the Euclidean distance between the adjacent β-nodes is larger than $r_\beta + c_2(\lambda_\alpha)$, all β-nodes in hybrid network $H(\alpha,\beta)$ are efficiently used with probability nearly 1. Therefore, when $\lambda_\beta = \frac{m}{d^2} < \frac{1}{(r_\beta + c_2(\lambda_\alpha))^2}$, all β-nodes in the network are efficiently used with probability nearly 1. □

Lemma 2 provides the sufficient condition that all β-nodes in hybrid network $H(\alpha,\beta)$ are efficiently used with probability nearly 1, for the case $r_\beta > 1$. Then, we have the following theorem which illustrates the impact of r_β on the connectivity of α-nodes.

Theorem 3: In hybrid network $H(\alpha,\beta)$, for $r_\beta = k_0 c_2(\lambda_\alpha) + 1$ ($k_0 \in \mathbb{N}$), if the β-node density satisfies $0 < \lambda_\beta < c_3(\lambda_\alpha)$, β-giant component size scales with λ_β as $|\mathcal{C}_{(\alpha,\beta)}| \geq \frac{K_0\eta(\lambda_\alpha)n}{\lambda_\alpha}\lambda_\beta + o(n)$ with probability nearly 1.

Proof: To prove this theorem, we first study a particular hybrid network $\overline{H}(\alpha,\beta)$, which is described as follows. Consider a hybrid network $H(\alpha,\beta)$ described previously for $r_\beta = 1$, a number of additional α-nodes are placed at some specific locations on circles $Circ(\beta_i, kc_2(\lambda_\alpha))$ centered at β_i ($i \in \{1,2,\cdots,m\}$) with radius $kc_2(\lambda_\alpha)$ ($k \in \{1,2,\cdots,k_0\}$). The locations of the additional α-nodes are illustrated in the following way: for given i and k, let $\alpha_{a1}, \alpha_{a2}, \cdots, \alpha_{aq}$ be all additional α-nodes located on $Circ(\beta_i, kc_2(\lambda_\alpha))$. Without loss of generality, we assume $\alpha_{a1}, \alpha_{a2}, \cdots, \alpha_{aq}$ are in clockwise direction. Then the location of α_{aj} ($j \in \{1,2,\cdots,q\}$) is at the intersection of $Circ(\beta_i, kc_2(\lambda_\alpha))$ and $Circ(\alpha_{a(j-1)}, c_2(\lambda_\alpha))$.

For example, for $k_0 = 1$, as depicted in Fig. 7, six additional α-nodes $\alpha_{a1}, \alpha_{a2}, \cdots, \alpha_{a6}$ are placed on the circle $Circ(\beta_i, c_2(\lambda_\alpha))$. The locations of α_{aj} ($j \in \{1,2,\cdots,6\}$) is at the intersection of $Circ(\beta_i, c_2(\lambda_\alpha))$ and $Circ(\alpha_{a(j-1)}, c_2(\lambda_\alpha))$. From the proof of Theorem 1, event E_1 that β_i and α_{aj} ($j \in \{1,2,\cdots,6\}$) cannot be connected with each other through the relay of α-nodes happens with probability nearly 1. Given that event E_1 happens, we denote $(k_i)_1, (k_i)_2, \cdots, (k_i)_6$ respectively as the number of α-nodes connected by each additional α-node "directly". Further, let $(k_i)_7$ be the number of α-nodes connected by β_i "directly". Obviously, in hybrid network $\overline{H}(\alpha,\beta)$, the number of α-nodes connected by β_i and additional six α-nodes "directly", is $\Sigma_{j=1}^7 (k_i)_j$. From our discussion in Section III-B, $(k_i)_j$

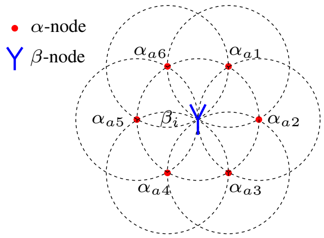


Fig. 7. The locations of additional α-nodes ($k_0 = 1$).

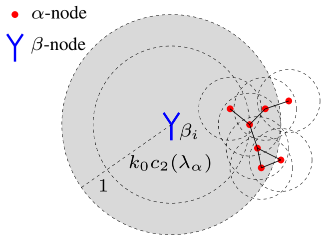


Fig. 8. The α-nodes connected by additional α-nodes placed on circle *Circ*$(\beta_i, k_0c_2(\lambda_\alpha))$ can be connected "directly" by β_i.

($j \in \{1, 2, \cdots, 7\}$) are i.i.d. random variables with finite mean $\eta(\lambda_\alpha)$ and finite variance. From the Weak Law of Large Numbers [11], the number of α-nodes connected by all β-nodes and all additional α-nodes is $7m\eta(\lambda_\alpha) + o(m)$ almost surely.

Then we consider the general case for any $k_0 \in \mathbb{N}$. Following the same logic as above, it is not difficult to know the number of α-nodes connected by all β-nodes and all additional α-nodes is $K_0m\eta(\lambda_\alpha)+o(m)$ with probability nearly 1, where $K_0 = 1 + \text{floor}(\frac{2\pi}{\arccos(\frac{1}{2})}) + \cdots + \text{floor}(\frac{2\pi}{\arccos(\frac{2k_0^2-1}{2k_0^2})})$ and floor(X) rounds X towards negative infinity.

Now, we consider hybrid network $H(\alpha, \beta)$ with the transmission range of β-nodes $r_\beta = k_0c_2(\lambda_\alpha) + 1$. For a given β-node β_i, it is not difficult to know that the α-nodes connected by all additional α-nodes placed on circles *Circ*$(\beta_i, kc_2(\lambda_\alpha))$ ($k \in \{1, 2, \cdots, k_0 - 1\}$) in $\overline{H}(\alpha, \beta)$, are within the transmission range of β_i and so can be connected "directly" by β_i. Then, we will show the α-nodes connected by additional α-nodes placed on circle *Circ*$(\beta_i, k_0c_2(\lambda_\alpha))$ can also be connected "directly" by β_i. From Fig. 8, it is obvious that the statement is true. Therefore, if the β-node density satisfies $\lambda_\beta < c_3(\lambda_\alpha)$, i.e., all β-nodes in the network are efficiently used with probability nearly 1, β-giant component size satisfies $|\mathcal{C}_{(\alpha,\beta)}| \geq K_0m\eta(\lambda_\alpha) + o(m) = \frac{K_0\eta(\lambda_\alpha)n}{\lambda_\alpha}\lambda_\beta + o(n)$ with probability nearly 1. □

For unit r_β, with a density of $\frac{1}{c_2^2(\lambda_\alpha)}$ β-nodes, β-giant component size can reach $S_1 = \frac{\eta(\lambda_\alpha)n}{\lambda_\alpha} \cdot \frac{1}{c_2^2(\lambda_\alpha)} + o(n)$ with probability nearly 1. For $r_\beta = k_0c_2(\lambda_\alpha)+1$ ($k_0 \in \mathbb{N}$), with a smaller density of $\frac{1}{(k_0c_2(\lambda_\alpha)+1+c_2(\lambda_\alpha))^2}$ β-nodes, β-giant component size can reach $S_2 = \frac{K_0\eta(\lambda_\alpha)n}{\lambda_\alpha} \cdot \frac{1}{(k_0c_2(\lambda_\alpha)+1+c_2(\lambda_\alpha))^2} + o(n)$. We plot the ratio $\frac{S_2}{S_1}$ for different α-node density in Fig. 9. As we can see from the figure, when k_0 increases from 1 to 40, the ratio keeps larger than 1 and approaches around 3, which demonstrates an additional benefit of enlarging r_β to improve the connectivity of α-nodes.

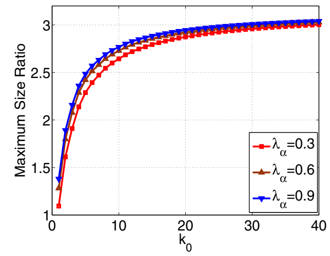


Fig. 9. The ratio of linearly achieved maximum β-giant component size.

IV. Conclusions

In this paper, we have studied how base stations in large-scale hybrid wireless network improve the connectivity of ad hoc nodes in subcritical phase. This problem is two-fold: the effect of density of base stations λ_β, and the effect of transmission range of base stations r_β on network connectivity. For unit r_β, we find the sufficient condition, i.e., $\lambda_\beta < c_1(\lambda_\alpha)$, that any two base stations cannot be connected through the relay of ad hoc nodes with probability nearly 1. Based on this result, we find that under $0 < \lambda_\beta < c_1(\lambda_\alpha)$, the number of connected ad hoc nodes is $\Theta(n)$ and scales linearly with λ_β with probability nearly 1. Further, we study the impact of transmission range of base stations r_β on network connectivity. We find out that by increasing r_β from unit to $k_0c_2(\lambda_\alpha) + 1$ ($k_0 \in \mathbb{N}$), with a smaller density of β-nodes, β-giant component can reach a larger size, which demonstrates an additional benefit of enlarging r_β to improve the connectivity of α-nodes.

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